



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

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Class : CSE
Batch : 2025-29

Assignment : 21

Subject: Programming with Java

ASSIGNMENT No : 21

TITLE : Demonstration and exploration of JDBC classes and interfaces (DriverManager, Connection, Statement).

Problem Statement

Demonstration and exploration of JDBC classes and interfaces (DriverManager, Connection, Statement).

Learning Objectives

To understand and use the core JDBC classes for database connectivity.

Theory

The JDBC API consists of several classes and interfaces that facilitate database connectivity. The DriverManager class is responsible for loading the appropriate JDBC driver and establishing a connection with the database using a database URL, username, and password. The Connection interface represents an active session between the Java application and the database server. It provides methods to create SQL statements and manage transactions. The Statement interface executes static SQL queries and updates. These classes work together to establish communication between Java programs and databases. Understanding these core components is essential for developing database applications using Java.



Conclusion : The JDBC classes and interfaces such as DriverManager, Connection, and Statement were successfully explored and demonstrated. DriverManager was used to establish a connection with the database, Connection managed the database connection, and Statement was used to execute SQL queries. Thus, JDBC provides an effective way for Java applications to interact with relational databases.

Exercise

Q1) Create a program to connect Java with database and display database connection status.

Sol) → SQL Queries :

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

#	Time	Action	Message	Duration / Fetch
1	22:50:51	CREATE DATABASE students	1 row(s) affected	0.015 sec
2	22:51:14	USE students	0 row(s) affected	0.000 sec



➔ JAVA Program :

```
J Main.java > Java > Main
1  import java.sql.*;
2
3  public class Main {
4      Run main | Debug main | Run | Debug
5      public static void main (String[] args) {
6
7          try {
8              Class.forName("com.mysql.cj.jdbc.Driver");
9
10             Connection con = DriverManager.getConnection(
11                 "jdbc:mysql://localhost:3306/students",
12                 "root",
13                 "Abcd@1234"
14             );
15
16             Statement stmt = con.createStatement();
17
18             System.out.println("Database connected successfully.");
19
20             con.close();
21         }
22         catch (Exception e) {
23             System.out.println("Connection failed: " + e.getMessage());
24         }
25     }
26 }
```

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```
PS C:\Users\LENOVO\Desktop\JDBC> javac -cp ".;lib\mysql-connector-j-26.7.0.jar" Main.java
PS C:\Users\LENOVO\Desktop\JDBC> java -cp ".;lib\mysql-connector-j-26.7.0.jar" Main
Database connected successfully.
PS C:\Users\LENOVO\Desktop\JDBC> 
```



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Q2) Create a Java program to connect with a database and display the connection status. After successful connection, show a message indicating that the Student database is connected successfully.

Ans) →SQL Queries :

The screenshot displays a SQL IDE interface. The main editor contains two queries:

```
1 CREATE DATABASE student;
2 USE student;
```

The right-hand pane shows a message: "Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help."

The bottom output window, titled "Output", shows the execution results for the queries:

#	Time	Action	Message	Duration / Fetch
✓ 1	22:50:51	CREATE DATABASE students	1 row(s) affected	0.015 sec
✓ 2	22:51:14	USE students	0 row(s) affected	0.000 sec
✓ 3	23:00:54	CREATE DATABASE student	1 row(s) affected	0.015 sec
✓ 4	23:00:54	USE student	0 row(s) affected	0.000 sec



➔ Java Code :

```
J StudentDatabase.java > Java > StudentDatabase
1  import java.sql.Connection;
2  import java.sql.DriverManager;
3
4  public class StudentDatabase {
5      Run main | Debug main | Run | Debug
6      public static void main(String[] args) {
7          try {
8              Class.forName("com.mysql.cj.jdbc.Driver");
9
10             Connection con = DriverManager.getConnection(
11                 "jdbc:mysql://localhost:3306/student",
12                 "root",
13                 "Abcd@1234"
14             );
15
16             System.out.println("Student database is connected successfully.");
17
18             con.close();
19         }
20         catch (Exception e) {
21             System.out.println("Student database connection failed.");
22             System.out.println(e.getMessage());
23         }
24     }
25 }
```

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```
PS C:\Users\LENOVO\Desktop\JDBC> javac -cp ".;lib\mysql-connector-j-26.7.0.jar" StudentDatabase.java
PS C:\Users\LENOVO\Desktop\JDBC> java -cp ".;lib\mysql-connector-j-26.7.0.jar" StudentDatabase
Student database is connected successfully.
PS C:\Users\LENOVO\Desktop\JDBC>
```